

Shubham Gupta

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Experience

IBM

Software Developer

Aug 2023 – Present

Bengaluru, India

- Built a Jira analytics dashboard using Python, Flask, and ReactJS (Carbon Design System), aggregating data via Jira API and deploying through a Docker-based CI/CD pipeline.
- Developed an **AI-driven support system** using Python, Ollama and Granite to automate Kubernetes cluster management. This system reduced manual intervention by intelligently retrying processes, resolving basic bugs, and managing job failures.
- **Architected a multi-source, multi-table data validation framework** in Python. Developed a configuration-based trend analysis tool using NumPy and Pandas to derive actionable insights from validated data.
- Optimized and maintained Apache Airflow DAG-based data delivery pipelines, **improving execution efficiency by 50%** and reducing runtime from 15 to 7.5 hours through pipeline optimization, debugging, and failure resolution.
- Built and maintained Apache Airflow DAG-based data validation workflows using UIF(unified ingestion framework), implementing SQL-driven YAML transformations for automated data processing and reporting.
- Leveraged IBM Bob AI assistant for intelligent code analysis, automated code review, and optimization across Python, JavaScript, and Shell scripting projects, improving code quality and development velocity.

IBM

Intern

Jan 2023 – Jul 2023

Bengaluru, India

- Developed a full-stack architecture diagram generator (ReactJS + Flask) that auto-creates system diagrams from Excel inputs, deployed on IBM Cloud.

Projects

Quantum Circuit Simulator

Apr 2025 – Sep 2025

- Built a web-based quantum circuit simulator using Qiskit AER, FastAPI, Python, and ReactJS
- Containerised the application using Docker for streamlined deployment and scalability.

RasqBerry Pi

Jan 2025 – Aug 2025

- Integrated quantum simulations and implemented installation checks using Python and Shell scripting to prevent redundancy and improve stability.

Technical Skills

Languages: Python, Shell Scripting, C++, JavaScript, SQL, Java

Frameworks & Tools: ReactJS (Carbon Design System), Node.js, Flask, FastAPI, Apache Airflow

Tools: IBM Bob (AI-assisted development), Ollama, Granite, Qiskit, RestAPI

Tools & Cloud: Git, Github, Docker, Postman, CyberDuck, DBeaver, Kubernetes, Linux, IBM Cloud

Education

Vellore Institute of Technology, Bhopal

2019 – 2023

B.Tech in Computer Science and Engineering

CERTIFICATIONS & AWARDS

IBM DevOps Fundamentals — IBM Agile Training — Industrial IoT Markets and Security